
SOFTWARE DEVELOPER INTERNSHIP REPORT

ARKENSYS REALTORS

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COURSE: B.E. Computer Science and Engineering
(First Year)

COLLEGE: SNS College of Technology, Coimbatore

REPORTING MANAGER: Mr. Harish Chakravarthy I (Managing
Director)

INTERNSHIP DURATION: 10 Days

TRAINING PERIOD: 18 June 2026 to 30 June 2026

Coimbatore, Tamil Nadu • Year 2026

ACKNOWLEDGEMENT

I thank Mr. Harish Chakravarthy I, Managing Director of Arkensys Realtors, for giving me the chance to work as a Software Developer Intern. I am grateful for his time, technical explanations, and direct guidance throughout these ten days.

I also thank Arkensys Realtors for providing me with a proper desktop workspace and access to real project environments. Working on actual tools helped me learn a lot of practical concepts that go beyond normal first-year classes.

COMPANY INTRODUCTION

Arkensys Realtors

Arkensys Realtors is a company that works in real estate development and property management. They plan and organize housing layout plots and residential properties for clients. The company builds its own custom internal software tools and runs practical technical projects. Instead of buying ready-made tools, they code their own applications to manage business needs. They also look into electronics, hardware circuits, and 3D printing to fix problems and learn new ways to manage systems.

WORKSPACE SETUP

On my first day, I was given my desk and a computer system to use for development work. My onboarding was quick and focused entirely on getting my system ready. I received my workstation, desktop computer, email access, and an introduction to the project assignment. I learned how to use the company's internal Task Tracker and began maintaining my own daily task logs to note down my features, progress, and errors.

DAY-WISE WORK AND LEARNING LOG

Day 1: 18 June 2026 – Project Assignment and Setup

Objective: Start the primary project and set up the coding files on my system.

Work Done: I was assigned to my main project, a web app called Artist Color Lab, which I built from scratch. I set up the project files using Vite with React and TypeScript. I used Antigravity, Claude, and ChatGPT throughout development, understood the code, tested everything, fixed errors, and ran the project on my local system. I checked through the files to see how the folders connect.

Learning: I learned how components link up in React and how TypeScript alerts you about mistakes while typing.

Tools Used: Antigravity, Claude, React, TypeScript, Vite.

Result: The base files were ready and running on my computer.

Day 2: 19 June 2026 – Landing Page Design

Objective: Build a clean, clear layout for the application homepage.

Work Done: I worked on the landing page layout for Artist Color Lab. I built the header bar, top informational section, and layout grid blocks using Tailwind CSS utilities. I used Antigravity, Claude, and ChatGPT throughout development to get the exact design classes needed to make the layout adapt nicely to phone screens. I changed the margins and background colors manually until it looked right.

Learning: I learned how Tailwind CSS helps you style a website quickly using built-in design shorthand instead of separate CSS files.

Tools Used: Claude, ChatGPT, Tailwind CSS, React.

Result: The homepage layout was designed and looked clean on both desktop and mobile windows.

Day 3: 20 June 2026 – Database Connection and User Login

Objective: Link the app to a database and set up a working login screen.

Work Done: I created an account on Supabase and made a new database project. I connected these keys to my local project code. Then I built a One-Time Password (OTP) user login feature. I used Antigravity, Claude, and ChatGPT throughout development, understood the code, tested everything, fixed errors, and managed the connection. When the system threw connection token errors, I read the logs and fixed the route path manually.

Learning: I learned how data moves between a frontend website and a cloud database, and how user sessions stay active through tokens.

Tools Used: ChatGPT, Antigravity, Supabase, React.

Result: Users can type their details and sign in using a functional OTP system.

Day 4: 22 June 2026 – Paint Library and Collections

Objective: Create database tables to let users save individual color groups.

Work Done: I went into the Supabase panel and set up relational tables to hold color tags, hex values, and user details. Then I wrote frontend code to display these entries in a simple list. I used Antigravity, Claude, and ChatGPT throughout development, understood the code, tested everything, fixed errors, and mapped out the functions so users could click a button to save custom colors into separate groups. I verified the entries inside the database tables manually.

Learning: I learned how tables store data rows using references and how to load that data into a web page using search commands.

Tools Used: Claude, Supabase, React.

Result: The custom library section worked, and color groups saved properly to the backend database tables.

Day 5: 23 June 2026 – Color Mixer and Camera Selection Features

Objective: Add functions to mix colors and read shades from images or live camera views.

Work Done: I implemented code logic to calculate the mix blend between colors. I also built an image upload sampler tool and a camera capture box that uses the browser canvas to read hex tags from photos or video. I included a gradient color picker tool as well. I used Antigravity, Claude, and ChatGPT throughout

development, understood the code, tested everything, fixed errors, and manually adjusted the controls to fit my UI elements.

Learning: I learned how browsers see images as arrays of pixels and how to ask for user camera permissions inside JavaScript code.

Tools Used: Claude, ChatGPT, Canvas API, React.

Result: The mixer, canvas camera sampler, and gradient elements all worked together without crashes.

Day 6: 24 June 2026 – Account Deletion and Site Policies

Objective: Create data wipe features, write static policy files, and prepare layout headers for site ads.

Work Done: I built an account deletion button that completely clears a user's profile rows and saved color lists from our database tables. I also created static text blocks for the app's standard legal pages, which include the Privacy Policy and Terms of Service. Finally, I placed basic verification tags in the site HTML head area to prepare the template for future Google AdSense validation updates. I used Antigravity, Claude, and ChatGPT throughout development to verify the setup.

Learning: I learned that you must give users a way to delete their personal accounts completely, and how structural tags work inside a website header.

Tools Used: ChatGPT, Supabase, React.

Result: Deletion options and basic policy pages were operational, and headers were ready for checking.

Day 7: 25 June 2026 – Hosting Live and Mapping Custom Domains

Objective: Deploy the web application live to the public internet using a clean address.

Work Done: I linked my code files to Cloudflare Pages to set up the live site. I ran the final build of the React app and cleared build errors that showed up. Then, I opened the company GoDaddy portal and shifted the DNS nameserver targets for entron.in over to Cloudflare. I used Antigravity, Claude, and ChatGPT throughout development, understood the code, tested everything, fixed errors, and deployed the application successfully.

Learning: I learned how global DNS changes take time to distribute and how domain records (like A and CNAME) point an address to specific hosting servers.

Tools Used: Antigravity, GoDaddy, Cloudflare Pages.

Result: Artist Color Lab went fully live and functional at the public link entron.in.

Day 8: 26 June 2026 – Project 2: Interactive Plot Map

Objective: Work on a new mapping visual for the company property pages.

Work Done: I worked on the Interactive Plot Map feature for the main Arkensys website. I took coordinate details from data charts stored in CSV files and mapped them out as visual SVG shapes. I built a hover feature that reveals plot measurements and prices when you move your mouse cursor over a shape block. I linked this to a separate new database in Supabase. The feature was still under development when my internship ended.

Learning: I learned how SVG layouts draw shapes on screen using numbers and how to change row text like CSV datasets into active shapes.

Tools Used: Claude, ChatGPT, SVG layouts, Supabase.

Result: A functional visual mockup was built that shows plot metrics whenever a user hovers over a shape boundary.

Day 9: 29 June 2026 – Practical Circuit Prototyping

Objective: Learn real electronics assembly and how sensors switch electricity lines.

Work Done: I went to the lab room to work with actual electronics components. I put together parts on a breadboard and tried out three sensor pieces: the KY-026 Flame Sensor, the KY-013 Temperature Sensor, and the KY-003 Hall Sensor. I set up a BC547 NPN transistor to turn a relay box on and off. I used a multimeter to measure voltage paths and check batteries. When my motor circuit failed to move, the Managing Director showed me that the battery current was too low to power the motor load. I drew out the circuit lines in my notes after understanding it.

Learning: I learned about Collector, Base, and Emitter lines, and the difference between Normally Open (NO) and Normally Closed (NC) terminals. I also learned about Common Ground lines and got a basic look at power switching parts, linking this back to my engineering physics class.

Tools Used: Breadboard, Sensors, BC547 Transistor, Relay, Multimeter, Batteries.

Result: Built a working flame sensor circuit and understood how a small signal can switch larger power paths.

Day 10: 30 June 2026 – Arduino Coding and Final Notes

Objective: Program a simple microcontroller board to handle components and finalize my logs.

Work Done: I spent my final session looking at an Arduino Uno board. I wrote a basic sketch containing the standard setup() and loop() code functions. I used commands like pinMode(), digitalWrite(), and delay() to change output signals. I uploaded the logic and got two LEDs to blink alternately. I am still a beginner in this field. I finished my task tracker entries, summarized my logs, and received my company internship merchandise.

Learning: I learned how software rules hardware output lines by toggling voltage and how microcontrollers process a continuous loop function.

Tools Used: Arduino IDE, Arduino Uno, LEDs, Jumper lines.

Result: Verified a basic alternate flashing dual-LED setup and completed all training steps.

PRACTICAL LEARNING AND PHYSICAL PROJECTS

Outside of web programming, Arkensys Realtors highlights hands-on building. This section outlines the mechanical and electronic systems I learned about.

1. 3D Printing

I studied the fundamental components and steps required to use a 3D printer. I learned how digital files are sliced into flat paths to build items layer by layer.

- **Printer Workflow:** I looked at the printer's X, Y, and Z axes, how the hot extruder melts material, and how the bed holds the piece. I used PrusaSlicer to process model layers.
- **Calibration and Prints:** I learned about bed leveling and the Z Offset spacing distance. My first actual test print failed completely because the melted plastic didn't stick to the bottom plate. Instead of ignoring the error, I checked the setup with my manager and learned that wrong height targets cause adhesion failures. Fixing this taught me exactly why print lines fail.

2. Electronics and Circuit Building

To better understand the formulas from my physics classes, I physically constructed circuits on breadboard strips.

- **Sensor Circuits:** I built systems using a KY-026 Flame Sensor to read fire signatures, a KY-013 Temperature Sensor to measure temperature changes, and a KY-003 Hall Sensor to detect magnetic items.
- **Transistors and Switching:** I used a BC547 NPN Transistor to control power lanes safely. I connected it to a mechanical relay box, learning how a low-voltage microcontroller pin can tell a relay to snap an internal metal plate and connect a completely separate high-voltage loop. I practiced splitting paths into Normally Open (NO) and Normally Closed (NC) lines and connected everything back to a Common Ground line. I also learned the core basics of why larger transistors are used for heavier power switching.

3. Arduino Programming

I learned the entry rules of programmable electronics by running code blocks on an Arduino Uno board.

- **Code Structure:** I learned that every board file requires a `setup()` function to set pin input/output directions and a `loop()` function to repeat actions indefinitely.
- **Commands Used:** I used basic commands like `pinMode()` to open pins and `digitalWrite()` to turn voltage HIGH or LOW. By putting these commands inside my code alongside specific `millisecond delay()` periods, I successfully programmed two separate LEDs to flash back and forth. This helped me understand the baseline basics of embedded code.

OTHER TECHNICAL OBSERVATIONS

During my ten days at Arkensys Realtors, I attended several informational discussions and saw equipment demonstrations conducted by the Managing Director.

Microscope Demonstration

I looked through a digital microscope setup to see highly detailed items, including a cardiac muscle sample and small plant insects. This showed me how high-zoom lenses capture clear data for analysis.

Solar Energy Basics

The Managing Director explained the physical components inside solar layouts. I learned how treated silicon materials create cells and how light particles hit those plates to cause electricity flow across wires.

Computer Motherboards

I got to look closely at a desktop motherboard component after opening up a computer cabinet casing. I saw where the processor sits, the RAM expansion tracks, cooling fans, power delivery lines, and hard drive connections. This let me see the actual hardware components that run our code.

Hosting and Deployment Choices

I had multiple informal technical conversations with Mr. Harish Chakravarthy I. I discussed why Cloudflare Pages was chosen for hosting and how using cloud hosting makes deployment easier. This helped me understand how project choices modify setup and load speeds.

WORK ENVIRONMENT AND RESOURCES PROVIDED

The company maintained a professional work environment by providing the necessary development resources, a dedicated workspace, access to software tools, and regular refreshments during the working day. This section details the complete logistical resources and tools that were provided by the company to enable my work.

Work Resources

For my daily assignments, Arkensys Realtors provided the following structural office assets:

- A dedicated workstation and physical workspace inside the company offices.
- A dedicated desktop computer system configured for software development activities.
- A formal company Google account and workspace account for secure access to company project environments.
- Access to the company's internal Task Tracker utility along with project documentation files.
- A systematic daily task logging system used to record features, progress updates, and code syntax bugs.

Software and AI Resources

To support an efficient engineering workflow, the company provided access to professional cloud platforms and AI tools. These utility tools were utilized responsibly as development assistants throughout the development of the projects:

- **AI Assistants:** Antigravity, Claude, ChatGPT, and Google AI Studio (Gemini). These tools were used to speed up feature implementation, code generation, debugging error logs, learning complex concepts, styling UI elements, and finding deployment assistance. AI assisted my work, but I personally read, understood, tested, modified, and verified the code structures before using them.
- **Collaboration Ecosystem:** Google Business Standard workspace, which enabled communication via official company email channels and team collaboration assets.
- **Infrastructure and Hosting:** Cloudflare Pages for publishing web projects, GoDaddy DNS consoles for custom domain management, and Supabase instances for database tables.

Learning Environment and Technical Exposure

The internship layout was built to support hand-on, practical learning rather than textbook theory memorization. This was driven by regular discussions with the Managing Director, Mr. Harish Chakravarthy I, which focused on active idea sharing and real-world problem solving.

Apart from building web databases, the company provided additional learning opportunities that exposed me to hardware setups. This included electronics assembly, circuit building on breadboards, 3D printing workflows, Arduino microcontroller basics, internal desktop hardware components, solar panels, and digital microscope demonstrations.

Workplace Environment and Internship Merchandise

To support daily operations, morning refreshments and evening refreshments were provided to employees and interns as a regular part of the daily work environment.

Upon the successful completion of the training blocks, the final sign-off process included receiving my official internship certificate alongside company merchandise, which consisted of an Arkensys T-shirt and an Arkensys cap. These resources are recorded here factually to document the complete professional environment provided to interns.

PROJECTS AND TRAINING SUMMARY

This summary table highlights the status, technical tools, and actual practical results of the projects and modules I worked on during my internship:

PROJECT / TOPIC	TECHNOLOGIES & ELEMENTS USED	STATUS	PRACTICAL RESULT ACHIEVED
Artist Color Lab	React, TypeScript, Tailwind CSS, Supabase, Cloudflare Pages, GoDaddy DNS	Completed	Built a full color web app from scratch and hosted it live at entron.in.
Arkensys Website	SVG Shapes, CSV Coordinate Data, Supabase Tables	Under Dev	I worked on the Interactive Plot Map feature. The feature was still under development when my internship ended.
3D Printing	PrusaSlicer, Filament Extrusion, Bed Leveling Coordinates	Completed	Learned printer assembly mechanics and how to calibrate bed leveling.
Electronics Prototyping	Breadboard, Sensors, BC547 Switching Transistor, Relay Blocks	Completed	Wired a functional flame sensor circuit and used common ground rules.
Arduino Programming	Arduino Uno Microcontroller, Pin Logic, Digital Output Controls	Completed	Wired and uploaded a program to blink two separate LEDs alternately.

CONCLUSION

This ten-day training at Arkensys Realtors provided me with a clear, realistic look at software developer tools and hardware prototyping. Building real platforms was highly beneficial for my studies.

I learned that using AI tools like Antigravity, Claude, and ChatGPT is a smart way to develop software, provided you take the time to understand what the code does, test every single layout view, fix error prompts, and manage your custom domain settings correctly. I built an entire web application from scratch and saw it go live. I also learned to troubleshoot hardware circuits and calibrate physical machinery.

The daily task tracking habits and technical rules I picked up will help me write better logic and understand engineering structures throughout my remaining years in college.

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